IJEP 20 (6) : 452 - 455

(Received on December 1, 1999)

Seasonal Variation of Water Quality in a Tropical Kalyani Reservoir, Near Tirupati

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This paper mainly deals with the seasonal variation of physical, chemical and biological quality of water in a tropical Kalyani reservoir near Tirupati. The result indicate that there is a wide seasonal variation in the quality of water. Colour, pH, total solids, alkalinity, dissolved oxygen, BOD and COD, bicarbonates, and chlorophyll 'a' concentrations fluctuate significantly in different seasons. These fluctuations are attributed by the influx of various organic and inorganic chemical constituents into the reservoir through run - off from the catchment area. This has also resulted propound impact on the chemical cost of water treatment in different seasons of the year.

INTRODUCTION

The development of water resources continues at a relatively rapid rate in India in response to the need for public water supply and other purposes. In view of increased water demand for the population of Tirupati, a fast urbanising pilgrim - centre in Andhra Pradesh, it became necessary to have a dependable sources of water supply to the public. Accordingly, the Tirupati Municipality constructed Kalyani Reservoir in 1977 at a cost of Rs. 143 crores across the river Kalyani a major tributary to Swarnamukhi river. This reservoir has a potential to supply 18.2 million L/day of water at a design period of 3 year.

Water quality and other ecological changes are generally unavoidable whenever a freely flowing straam and its riperian zones are committed to permanent inundation in the form of a reservoir (Swenson and Baldwin, 1965). The quality of impounded water is subjected to major physical, chemical and biological changes due to the influx of sediments and dissolved substances from the catchment area with which the water comes in contact with rocks, soils, and vegetation during run - off. The reservoir water is generally composed of surface run - off and ground water and the relative amounts of these two components are constantly changing owing chiefly to change in the rates of precipitation and evapotranspiration. The dissolved and suspended soils contained in the reservoir water are derived from all materials with which the water comes in contact, including the atmosphere, vegetation, soil ana rock. A significant variation of water quality in the reservoir can be expected due to the nature of input of water from these sources of contamination seasonally. This will have a propound impact on the treatment processes of the water. The major objective of this investigation is to characterize - (1) the seasonal variation of water quality in the reservoir of Kalyani (2) calculation of annual transport of different chemical constituents into the reservoir from catchment area and (3) associated cost of water treatment due to the seasonal variation of water quality during the year.

Description of the reservoir

Kalyani Reservoir is located at distance of 20 km from Tirupati at a longitude 79°16′ and latitude 13°39′. The important rivulets that join Kalyani river are Saddikudu Vanka, Kukhala Kona Vanka, Anupukona, Ragimanu Kona, Mamidimanu Kona, Pagadagundla Kona, Chintagunta Vanka and Negapatla Vanka. All the above rivulets are ephemeral in nature. The basin forms a part of the Mysore Plateau of Eastern Ghats of undifferential crystallines covered by granites agd quartzite and has an elevation ranging from + 378 m to + 914 m. The major portion of the drainage basin is covered with hill ranges intercepted by narrow valleys and forest areas and 85 % of the area is under more than 14° slope.

The drainage area is dominated by the general forest with various types of vegetation. Ramachandraiah (1986) had identified 78 tree species in the drainage area. Due to the excessive slopes and pre-

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	(July to Sept)	soon (Oct to Nov)	to March)	(April to June)
Colour	Muddy brown	Light brown	Light green	Deep green
Taste and odour	Soil odour	Swampy taste	Swampy taste	Algal odour
pH	7.40	7.15	7.60	7.80
SEC, μ mhos/cm	360	290	265	280
Turbidity, JTU	100	40	30	60
Total solids	610	390	210	265
Dissolved solids	220	205	160	170
Suspended solids	390	85	150	95
Total acidity as $CaCO_3$	8	10	10	5
Total alkalinity as $CaCO_3$	110	95	60	90
Total hardness as $CaCO_3$	85	66	65	60
Dissolved oxygen	5.3	6.6	8.8	7.8
Sodium as Na^+	28	26	26	20
Calcium as Ca^+	18	15	16	12
Magnesium as Mg^{++}	9.8	6.8	6.5	7.3
Iron as Fe	0.28	0.16	0.15	0.11
Chlorides as Cl^-	12	10	11	12
Sulphates as SO_4^{-2}	18	12	10	10
Silica as SiO_2	10.5	8.3	8.1	4.1
Bycarbonates	136	[illegible]	[illegible]	[illegible]

nt area

Parameter	Average concentration, tonne/year
Dissolved solids	375.00
Suspended solids	2215.00
Sodium as Na^{++}	112.50
Calcium as Ca^{++}	56.25
Magnesium as Mg^{++}	37.50
Chlorides as Cl^{-}	12.50
Sulhates as SO_4^{-2}	50.00
Silica as SiO_2	21.25
Soluble BOD_5 at 20 °C	143.75
Soluble COD	337.50

Table 4. *Optimum dose of alum for different seasons of the reservoir water*

Season	Lime, mg/L	Dose of alum, mg/L	Treatment cost of water, Rs. per m^3
Monsoon (July to September)	30-40	50-60	5.40-6.10
Post-monsoon (October to November)	10-20	30-40	3.45-3.90
Winter (December to March)	nil	15-20	1.10-14.5

more during monsoon due to the higher turbidity of water. Lime addition to raw water prior to coagulation is needed during monsoon and post - monsoon periods.

CONCLUSION

There is a wide seasonal variation of water quality in the Kalyani reservoir. The colour of the reservoir water mostly light green to deep green from October to June while muddy brown to light brown during the rest of the year. The turbid condition of the reservoir during monsoon was a result of transport of various organic and inorganic chemical constituents from the catchment area through run - off. The cationic sequence of the water is $Na^{+} > Ca^{++} > Mg^{++}$ and the anion sequence is $HCO_3^{-} > SO_4^{-2} > Cl^{-}$. The cost of treatment of water during monsoon is higher due to excessive lime and alum doses require for coagulation, flocculation of reservoir water.

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KEYWORDS

Water quality, Phytoplankton, Coagulation, Flocc lation, Kalyani reservoir.